In this document, we show that, if we take n samples from $\mathcal{N}(\mu, \sigma^2)$, the minimum distance between two samples is $\Theta\left(\frac{\sigma}{n^2}\right)$ with high probability.

First, we draw $\frac{n}{2}$ samples. If a pair of these samples are within a distance $\frac{c\sigma}{n^2}$, we are done. Otherwise, we draw an interval of width $\frac{2c\sigma}{n^2}$ centered around each point which is within one standard deviation of the mean. We note that, the expected number of points which fall within one standard deviation of the mean is $\frac{n}{2} erf\left(\frac{1}{\sqrt{2}}\right)$. Using Markov's inequality on the complement of this random variable, we can verify that the probability that we have fewer than $\frac{n}{4} erf\left(\frac{1}{\sqrt{2}}\right)$ is $\leq \frac{1}{2}$. Furthermore, we note that the minimum value of the pdf in these intervals will be at $\mu \pm \sigma$, which is the value $\frac{1}{\sigma\sqrt{2\pi e}}$.

We can find a lower bound for the probability measure of these intervals. Each one is of length $\frac{2c\sigma}{n^2}$. The PDF within each interval is at least $\frac{1}{\sigma\sqrt{2\pi e}}$. With probability at least $\frac{1}{2}$, we have $\frac{n}{4} erf\left(\frac{1}{\sqrt{2}}\right)$ such intervals. Multiplying these all together, we get $\frac{c erf\left(\frac{1}{\sqrt{2}}\right)}{2n\sqrt{2\pi e}}$.

Now, we draw the remaining $\frac{n}{2}$ samples. The probability that a sample lands outside these intervals is $1 - \frac{c erf\left(\frac{1}{\sqrt{2}}\right)}{2n\sqrt{2\pi e}}$. The probability that all samples land outside these intervals is $\left(1 - \frac{c erf\left(\frac{1}{\sqrt{2}}\right)}{2n\sqrt{2\pi e}}\right)^{\frac{n}{2}} \leq exp\left(-\frac{c erf\left(\frac{1}{\sqrt{2}}\right)}{4\sqrt{2\pi e}}\right)$. The probability that the minimum distance is less than $\frac{c\sigma}{n^2}$ is $\geq 1 - exp\left(-\frac{c erf\left(\frac{1}{\sqrt{2}}\right)}{4\sqrt{2\pi e}}\right)$.

Therefore, with high probability, the minimum distance between two samples is $O\left(\frac{\sigma}{n^2}\right)$.

In a similar way, we give a lower bound, showing that the minimum distance between two samples is $\Omega\left(\frac{\sigma}{n^2}\right)$. Consider the complement of this event - where no pair is within a distance $\frac{c\sigma}{n^2}$. If none of the first j points are within a distance $\frac{c\sigma}{n^2}$, we can draw an interval of width $\frac{2c\sigma}{n^2}$ centered at each of the points. Since the maximum of the PDF is $\frac{1}{\sigma\sqrt{2\pi}}$, the probability measure of these intervals is $\leq j\frac{2c\sigma}{n^2}\frac{1}{\sigma\sqrt{2\pi}} \leq n\frac{2c\sigma}{n^2}\frac{1}{\sigma\sqrt{2\pi}} \leq \frac{2c}{n\sqrt{2\pi}}$. The probability of the $j+1$st point not falling in any of these intervals is at $\geq 1-\frac{2c}{n\sqrt{2\pi}}$. If we find the probability that each point falls outside the intervals, given that the previous points fell outside the intervals, we have the probability that the minimum distance is at least $\frac{c\sigma}{n^2}$ is $\geq \left(1-\frac{2c}{n\sqrt{2\pi}}\right)^n$. Taking the complement of this event, we see the probability that there exists a pair of points within a distance $\frac{c\sigma}{n^2}$ is $\leq 1-\left(1-\frac{2c}{n\sqrt{2\pi}}\right)^n \leq 1-\left(1-\frac{2c}{n\sqrt{2\pi}}\right)^{\frac{2c}{\sqrt{2\pi}}} e^{-\frac{2c}{\sqrt{2\pi}}}$.